

# Assignment 3 Types, Diagrams, and Deductions

---

Copyright (c) 2026 Chris Liu, Dustin Tucker, James B. Wilson All rights reserved.  
Not permitted for AI training nor public posting.

## 1.

---

Using the following syntax of logical OR ( $\vee$ ), create an associated data type using F.I.C.E. rules (you may assume all contexts are implicit for simplicity).

$$\frac{P}{P \vee Q} (I_{*\vee}) \quad \frac{Q}{P \vee Q} (I_{\vee*}) \quad \frac{\begin{array}{c} P \vee Q \\ P \rightarrow R \\ Q \rightarrow R \end{array}}{R} (E_{\vee})$$

- A. Formation
- B. Introduction
- C. Elimination
- D. Computation

## 2

---

In the following programs, identify if it is likely to be a “Formation”, “Introduction”, “Computation”, or “Elimination” rule. Give a one line reason why.

1. `new File("text.txt")`
2. `public class Matrix<K>`
3. `u = x.name`
4. `Link(tag).getTag() == tag`
5. `age where user=User(name,age)`
6. `makePoint 4 5`

7. interface Serializable
8. extract(p)
9. save(f)
10. obj.toString()

### 3.

---

Convert the following program into a sentence in logic using sequents. (We are expecting an implication as the answer)

```
def plot(data:DataFrame, row1:Int, row2:Int):Plotter
```

### 4.

---

Given the logical sentence, list the data type/program in pseudo-code that would match the logic.

If P then (Q and R leads to S)

### 5.

---

You are asked to create a “print” command to take in strings as input. You are told you can also use “Void” (a data type with no values), a Boolean (a data type with 2 values), or you can raise exceptions. You are told that the print command almost never fails and certainly the users won't be checking on its status.

Rank the strength/weakness of each of the following type choices to match the logic just described.

```
print1 : String -> (Void+Exception)
print2 : String -> Boolean
print3 : String -> Void
```

## 6

---

Draw the commutative diagrams for the data type given by the following rules.

- Formation  $\frac{A, B}{A \sqcup B}$
- Introductions  $\frac{a:A}{\iota_A(a):A \sqcup B}$
- Introductions  $\frac{b:B}{\iota_B(b):A \sqcup B}$
- Elimination

$$\frac{x : A \sqcup B, f : A \rightarrow C, g : B \rightarrow C}{(f \sqcup g)(x) : C}$$

- Computation

$$\frac{a : A, f : A \rightarrow C, g : B \rightarrow C}{(f \sqcup g)(\iota_A(a)) = f(a)}$$

- Computation

$$\frac{b : B, f : A \rightarrow C, g : B \rightarrow C}{(f \sqcup g)(\iota_B(b)) = g(b)}$$

## 7

---

Given the following four commutative diagrams identify match each diagram to “formation”, “introduction”, “elimination”, and “computation rules” of the type.

Apple

Milk

